Sean Diab

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Research Engineer | ML Engineer | Bay Area, CA

Research engineer focused on language model architecture and evaluation; solo-built Reviser and FaceOff end-to-end.

Research & Projects

Reviser: Revision-Capable Language Modeling (Solo Research Project) GitHub | Paper Dec 2025 – Apr 2026

- Built and trained Reviser, a non-autoregressive language model that generates by editing a mutable canvas via autoregressive edit actions (Insert/Move/Stop); authored and published the accompanying paper
- Built a 1v1 arena evaluation framework with an LLM grader and ran 3k-sample C4 comparisons; Reviser achieved 61.3% (100M) and 54.4% (300M) win rate vs matched AR baselines, and 85.9%/78.3% vs SEDD/MDLM.
- Introduced and designed an obfuscation-to-restoration supervision pipeline by generating obfuscation trajectories and inverting them into restoration trajectories to produce scalable, edit-rich training data with strong revision signal
- Introduced and implemented canvas-state conditioning with a pooled canvas representation at each decoding step, efficiently improving state awareness with low overhead and without full-sequence recomputation.
- Built trajectory visualization and diagnostics confirming genuine revision during generation: 100% backward revisions, 96.1%/96.9% non-end appends (100M/300M), and 1.51/1.40 actions per token.
- Filed U.S. Provisional Patent Application covering core Reviser architecture and training methods.

FaceOff – Real-Time Streaming + Opponent Voice Platform for Clash Royale Jan 2026 – Jul 2026

- Solo-built and shipped a production platform (native SwiftUI iOS + Jetpack Compose Android apps, Python backend): live screen-share streaming with TikTok-style watching, automatic opponent detection connecting matched players in live voice chat, per-match replay clips, a ranked short-video community feed, and DMs/voice calls.
- Built the computer-vision matchmaking pipeline (masked template matching + RapidOCR over ReplayKit/MediaProjection screen capture) that recognizes two users in the same match and auto-connects them in a LiveKit voice room; cut cross-device detection skew from 14 s to sub-second via freshest-frame scheduling and threaded detection.
- Engineered the real-time media stack: 720p/60 fps hardware H.264 screen share with a simulcast fallback over LiveKit/WebRTC (replaced software VP9 after measured frame-rate drops), zero-re-encode track-egress recording with an ffmpeg CFR fallback, and replay clipping anchored to Supercell API battle timestamps.
- Implemented and deployed the full backend (FastAPI, PostgreSQL/asyncpg/Alembic, APNs/VoIP + FCM push, Apple/Google sign-in, moderation, rate limiting) via systemd and Cloudflare Tunnel with a dedicated LiveKit SFU and incremental R2 database backups; tested with pytest, contract-pinning XCTest/JUnit suites, and Maestro flows.
- Automated Clash Royale account verification end-to-end (users link accounts by playing one friendly match against a verifier bot) with OpenCV template matching and QR decoding on a GPU-passthrough Windows VM worker.

GPT Transformer Paper – Solo Research Project GitHub | March 2025 – June 2025

- Authored a comprehensive technical analysis of GPT forward/backward passes with full gradient derivations for backpropagation, including implementation-level checks that connect theory to practical training behavior.

Education

CU Boulder May 2024 – March 2025
M.S. Computer Science (completed before B.S.); Graduate Certificate in AI GPA: 4.0/4.0
Indiana University June 2025 – Dec 2025
B.S. Mathematics; Pi Mu Epsilon (Mathematics Honor Society) GPA: 4.0/4.0
College of San Mateo – Studied foundational STEM coursework Aug 2023 – May 2024

Skills

- Languages: Python, Swift, Kotlin, SQL, LaTeX
- ML/LLM: PyTorch, Transformers, NumPy, tokenization, training/inference pipelines, cloud GPU compute (Vast.ai)
- Backend/Systems: FastAPI, PostgreSQL (asyncpg/Alembic), LiveKit/WebRTC, ffmpeg, Linux/systemd, Docker
- Vision/Mobile: OpenCV, Pillow, RapidOCR, SwiftUI, ReplayKit, CallKit, Jetpack Compose, MediaProjection
- Evaluation & Testing: LLM-judge arenas, evalPPL workflows, benchmark analysis; pytest, XCTest, JUnit, Maestro

Interests: Chess (2200 rating), Clash Royale (top-100 global peak), long-distance cycling and running